

Mnl theory assignment 5

5.15

We need to compute the 2 integrals

$f(x) = x^a$ is continuous on any closed interval $[a, b]$ with $0 < a \leq b$

I denote this fact as (*)

a) The restriction of f to $(0, K]$ $\bar{f}$ is continuous and non negative function thus $\bar{f} \in \mathcal{M}(B(\mathbb{R}))^+$

We note that $f 1_{[\frac{1}{n}, K]}$ is a sequence of functions $f_1 \leq f_2 \leq \dots$

With $\bar{f} = \lim_{n \rightarrow \infty} f_n$ thus we can use theorem 5.2.4

$$\int_0^K x^a \lambda(dx) = \int \lim_{n \rightarrow \infty} f 1_{[\frac{1}{n}, K]} d\lambda = \lim_{n \rightarrow \infty} \int f 1_{[\frac{1}{n}, K]} d\lambda$$

But since f is continuous on the closed interval we get a sequence of Riemann integrals

$$I_n = \int f 1_{[\frac{1}{n}, K]} d\lambda = R \int_{\frac{1}{n}}^K x^a dx$$

This splits for different values of a

$$= \begin{cases} \left[\frac{x^{a+1}}{a+1} \right]_{\frac{1}{n}}^K & \text{for } a \neq -1 \\ [\log(x)]_{\frac{1}{n}}^K & \text{for } a = -1 \end{cases}$$

We start by looking at the case $a \neq -1$

$$\left[\frac{x^{a+1}}{a+1} \right]_{\frac{1}{n}}^K = \frac{K^{a+1}}{a+1} - \frac{\left(\frac{1}{n}\right)^{a+1}}{a+1}$$

Again the limit of this will depend on a

We start with $a < -1$

$$\frac{K^{a+1}}{a+1} \text{ is just constant}$$

$$-\frac{\left(\frac{1}{n}\right)^{a+1}}{a+1} \rightarrow \infty \text{ for } n \rightarrow \infty$$

Since $-\frac{1}{a+1} > 0$, $\left(\frac{1}{n}\right)^{a+1} \rightarrow \infty$ for $n \rightarrow \infty$

Meaning that we will have $I_n \rightarrow \infty$ for $n \rightarrow \infty$ for $a < -1$

Now $a > -1$

$$\left[\frac{x^{a+1}}{a+1} \right]_{\frac{1}{n}}^K = \frac{K^{a+1}}{a+1} - \frac{\left(\frac{1}{n}\right)^{a+1}}{a+1}$$

$$\frac{\left(\frac{1}{n}\right)^{a+1}}{a+1} \rightarrow 0 \text{ for } n \rightarrow \infty$$

Thus $I_n \rightarrow \frac{K^{a+1}}{a+1}$ for $n \rightarrow \infty$

For $a = -1$

$$I_n = \log(K) - \log\left(\frac{1}{n}\right) = \log(K) + \log(n) \rightarrow \infty \text{ for } n \rightarrow \infty$$

Thus we get

$$\int_0^K x^a \lambda(dx) = \begin{cases} \infty & \text{for } a \leq -1 \\ \frac{K^{a+1}}{a+1} & \text{for } a > -1 \end{cases}$$

b)

$$\int_K^\infty x^a \lambda(dx)$$

By an analogous argument as the one made in part 1 since f is cont and non negative on $[K, \infty)$ for any a we can rewrite

$$\begin{aligned} \int_K^\infty x^a \lambda(dx) &= \int \lim_{n \rightarrow \infty} f 1_{[K,n]} d\lambda = \lim_{n \rightarrow \infty} \int f 1_{[K,n]} d\lambda \\ I_n &= \int f 1_{[K,n]} d\lambda = R \int_K^n x^a dx \\ &= \begin{cases} \left[\frac{x^{a+1}}{a+1} \right]_K^n & \text{for } a \neq -1 \\ [\log(x)]_K^n & \text{for } a = -1 \end{cases} \end{aligned}$$

First $a < -1$

$$\begin{aligned} I_n &= \left[\frac{x^{a+1}}{a+1} \right]_K^n = \frac{n^{a+1}}{a+1} - \frac{K^{a+1}}{a+1} \\ &\text{since } \frac{n^{a+1}}{a+1} \rightarrow 0 \text{ for } n \rightarrow \infty \\ I_n &\rightarrow -\frac{K^{a+1}}{a+1} \text{ for } n \rightarrow \infty \end{aligned}$$

For $a = -1$

$$I_n = \log(n) - \log(K) \rightarrow \infty \text{ for } n \rightarrow \infty$$

For $a > -1$

$$I_n = \left[\frac{x^{a+1}}{a+1} \right]_K^n = \frac{n^{a+1}}{a+1} - \frac{K^{a+1}}{a+1}$$

Since $\frac{n^{a+1}}{a+1} \rightarrow \infty$ and the rest is constant

$$I_n \rightarrow \infty \text{ for } n \rightarrow \infty$$

Thus we get

$$\int_K^\infty x^a \lambda(dx) = \begin{cases} \infty & \text{for } a \geq -1 \\ \frac{K^{a+1}}{a+1} & \text{for } a < -1 \end{cases}$$